

Weekly Progress Report (07/28)

Topic : (Pending)

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Outline

1 、 Recently work

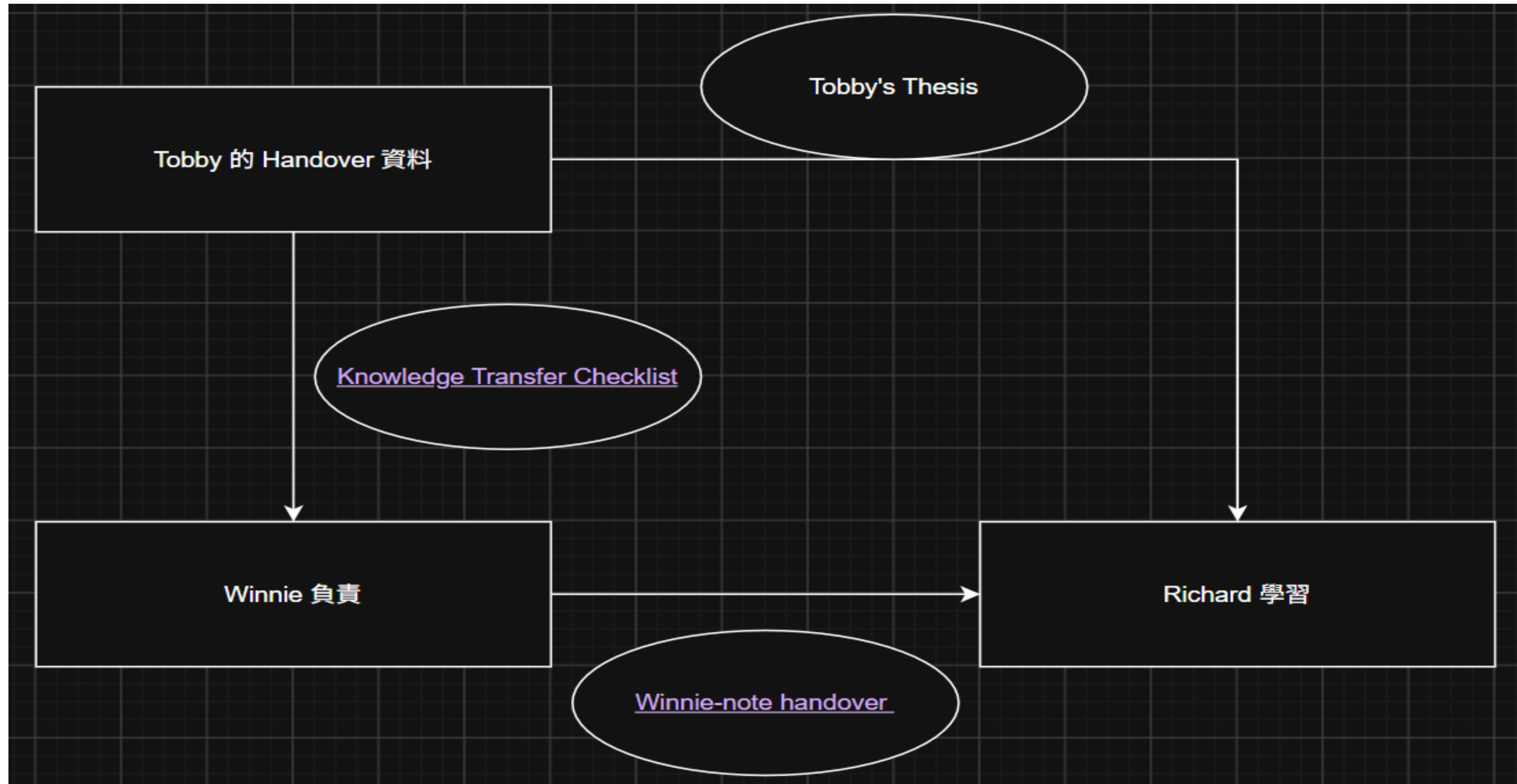
- > Replicate Toby's experiment
- > Reach out to TEEP David and time schedule

2 、 Pending work

- > Determine my topic on energy saving

Replicate Toby's experiment

Replicate Toby's experiment - 1



Replicate Toby's experiment - 2

Winnie note - handover	Why
Clone Toby's repository	Generate the experiment graph Private access requires Git to download
Environment Installation	Installing the Python extension in VS Code doesn't mean the system can run Python. Version problem cause to bugs
Check the cloned repository	Clarify the path 99% of the work is in Data_modeling
Check access to the lab network	VAIVA simulator isn't a public website. It requires the internal lab network.

Replicate Toby's experiment - 3

Problem	Solution
Locate the file path	cd, ls , dir ... Turn to AI for help
Link to VAIVA	1 、 Check whether VAIVA is alive. -> ping 192.168.8.28 2 、 VPN or SSH
Internal Internet	Turn to Joshevan for help Get a file (This is yours. Never share it with others.) , Download WireGuard
Check the VAIVA timetable	Turn to Toby for help.

Replicate Toby's experiment - 4

1 、 According to Toby's "Readme", Start running the experiment 1

- > Fix some bugs

- > Check the timetable

- > Stop the program if necessary

- > Compare with the expected result

Replicate Toby's experiment - 5

1、UnicodeDecodeError

-> This was caused by a Chinese path.

```
PS C:\Users\徐銘亨\intent-configuration-system\Data_modeling> cd C:\Users\徐銘亨\intent-configuration-system\Data_modeling
PS C:\Users\徐銘亨\intent-configuration-system\Data_modeling> python main.py
Traceback (most recent call last):
  File "C:\Users\徐銘亨\intent-configuration-system\Data_modeling\main.py", line 30, in <module>
    setup_config = yaml.safe_load(file)
  File "C:\Users\徐銘亨\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\yaml\__init__.py", line 125, in safe_load
    return load(stream, SafeLoader)
  File "C:\Users\徐銘亨\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\yaml\__init__.py", line 79, in load
    loader = Loader(stream)
  File "C:\Users\徐銘亨\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\yaml\loader.py", line 34, in __init__
    Reader.__init__(self, stream)
    ~~~~~^~~~~~
  File "C:\Users\徐銘亨\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\yaml\reader.py", line 85, in __init__
    self.determine_encoding()
    ~~~~~^~~~~~
  File "C:\Users\徐銘亨\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\yaml\reader.py", line 124, in determine_encoding
    self.update_raw()
    ~~~~~^~~~~~
  File "C:\Users\徐銘亨\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\yaml\reader.py", line 178, in update_raw
    data = self.stream.read(size)
UnicodeDecodeError: 'cp950' codec can't decode byte 0xe2 in position 1591: illegal multibyte sequence
PS C:\Users\徐銘亨\intent-configuration-system\Data_modeling> 
```


Replicate Toby's experiment - 6

2 、 Fix UnicodeDecodeError

- > Open main.py by IDE
- > In option manu, select “view the line”
- > Fix the code to the line.

Replicate Toby's experiment - 7

- Original

```
with open('config/config.yaml') as file:  
    setup_config = yaml.safe_load(file)
```

- adjusted

```
with open('config/config.yaml', encoding='utf-8') as file:  
    setup_config = yaml.safe_load(file)
```

Replicate Toby's experiment - 8

1、Check timetable

- > At the same time, only one person can use the simulator
- > When encountering bugs, stop the program in the terminal (Ctrl +C), then the environment will be reset, which prevent all events from entering a loop.

Replicate Toby's experiment - 9

1 、 compare the result (same_area_uniform)

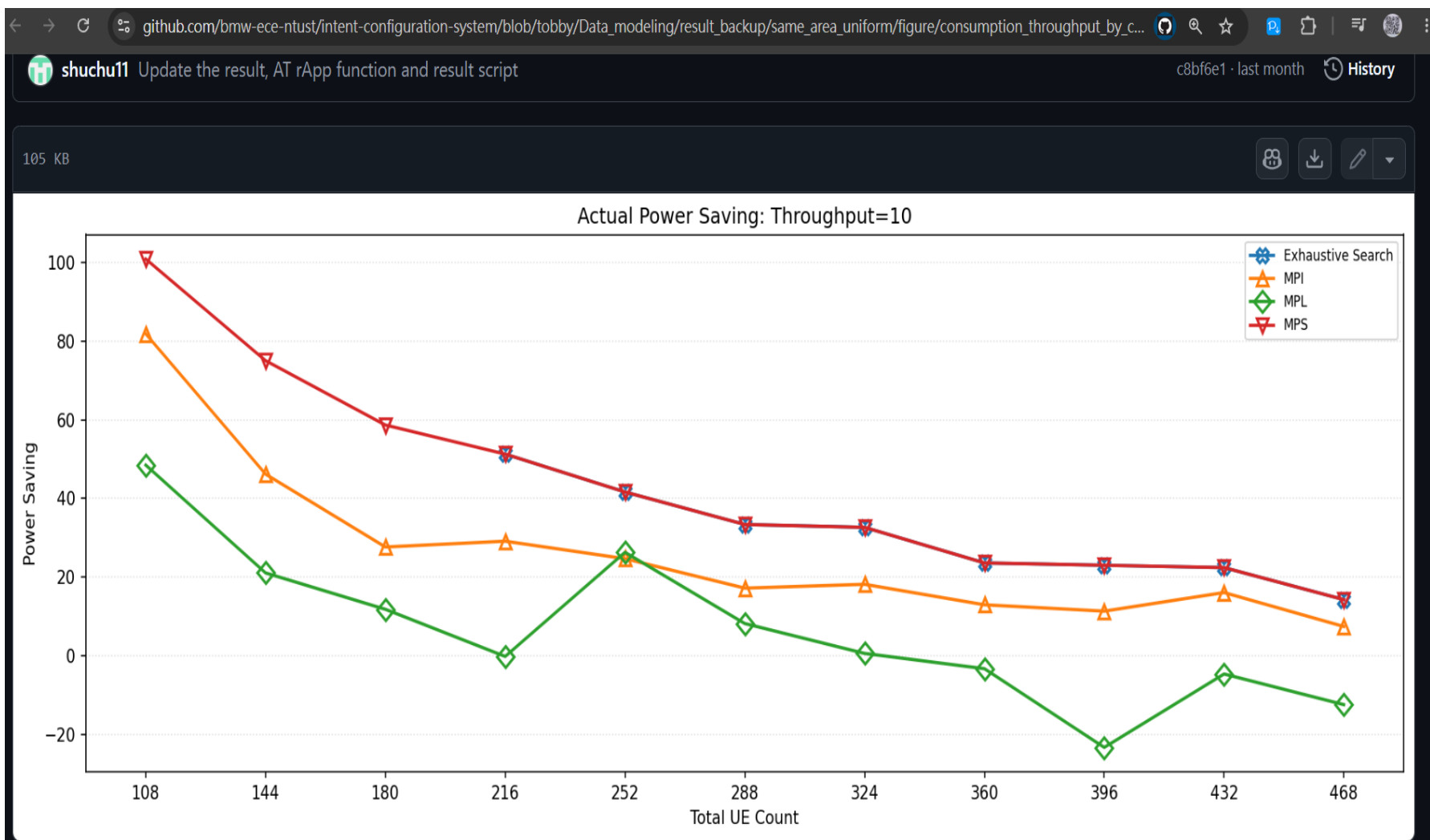
-> I can't view the picture through VS code, so open in folder directly

-> Compare the file name carefully

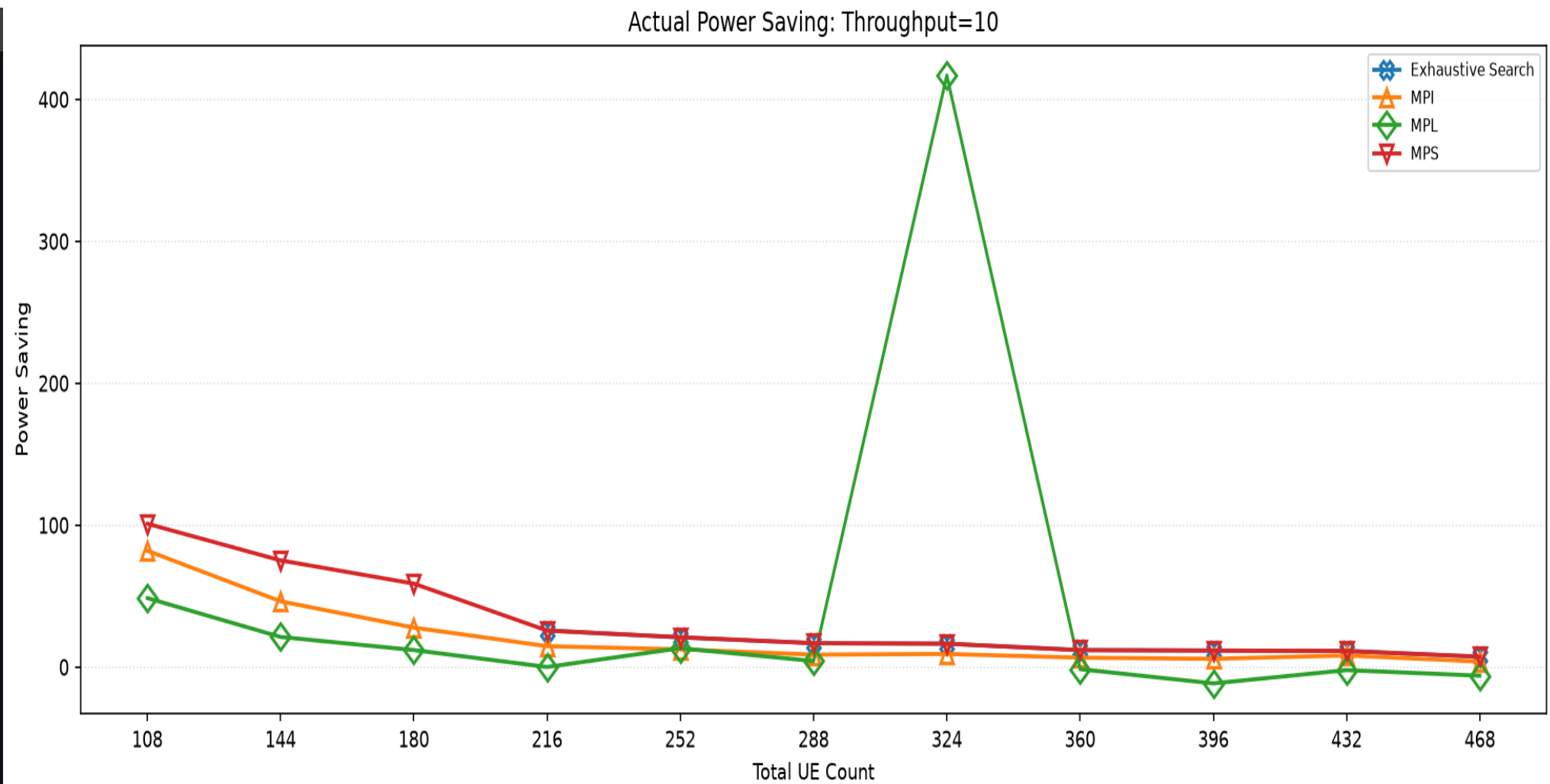
-> Understand the meaning of the result

Replicate Toby's experiment - 10

- Expected



- Mine



Replicate Toby's experiment - 11

2 、 Explain the abnormal result (same_area_uniform)

-> Experiment I (strategyTestMode: True), verify the algorithm

-> The abnormal power-saving value was caused by missing InfluxDB power data (CurrentPower = 0)

[illegible]

Replicate Toby's experiment - 12

1 、 compare the result (PMdiff_uniform)

-> I can't view the picture through VS code, so open in folder directly

-> Compare the file name carefully

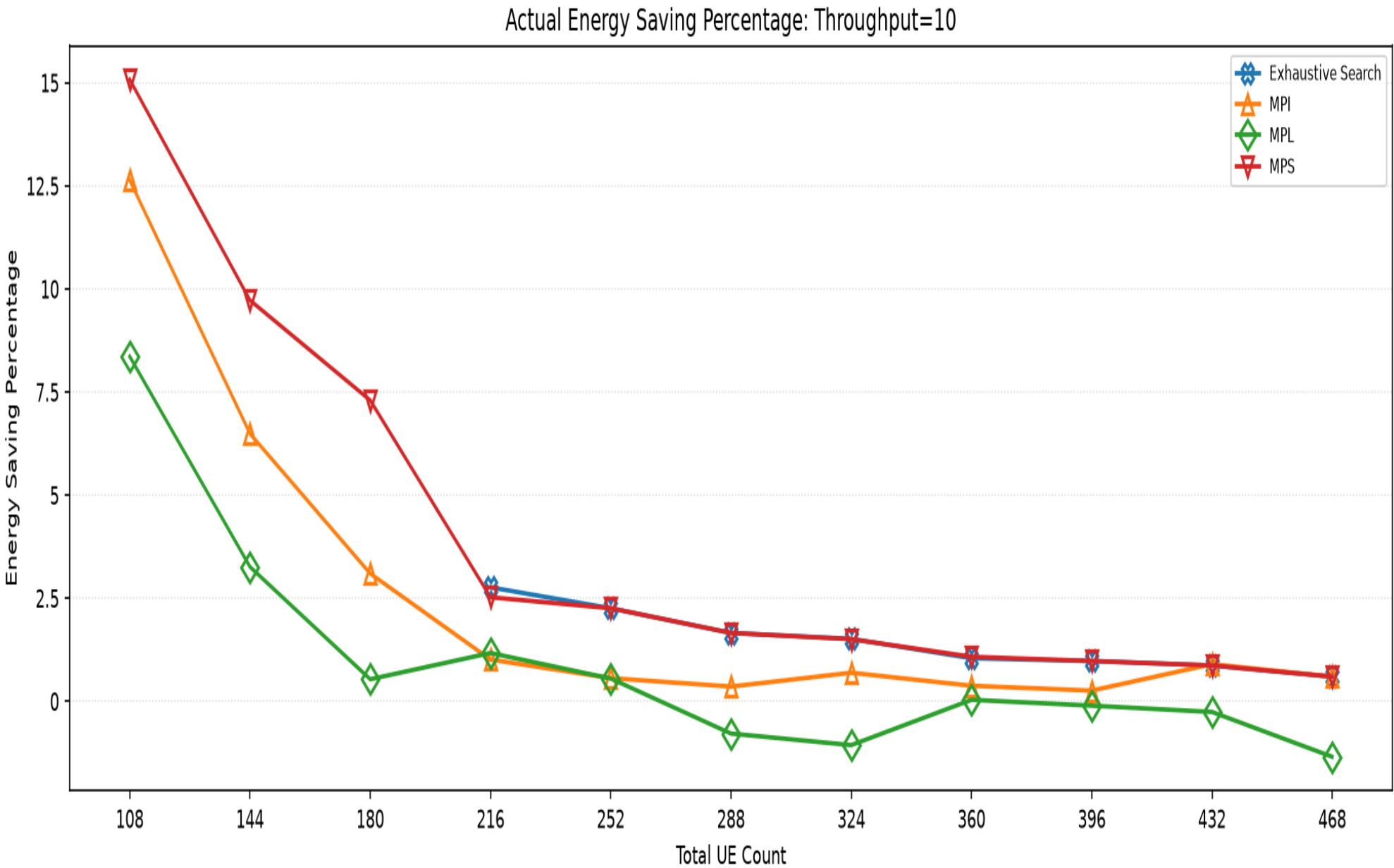
-> Understand the meaning of the result

Replicate Toby's experiment - 13

- Expected



- Mine



Replicate Toby's experiment - 14

2 、 Explain the result(PMdiff_uniform)

- > The trend is identical to expected result, some slightly differences are caused by the the simulator's randomness
- > The “uniform” means each UE's power consumption is the same, the independent variable is the area.
- > The former is UEs in the same area, the latter is UEs in different area.

Replicate Toby's experiment - 15

1 、 Next step

- > Timetable is reserved by the seniors
- > The simulation takes a lot of time (about 5hrs in generating data)
- > Reference Toby's result and draw conclusions directly, as background knowledge

Replicate Toby's experiment - 16

1、Timetable is reserved by the seniors

RSGReservation.XLSX

☆📎☁

轉換為PDF

共用

檔案 編輯 查看 插入 格式 資料 工具 Gemini 說明

選單 100% NT\$ % 123 Arial 11 B I A

R19fx

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1		Version: 2.6-4446																				
2																						
3		Date	Start Time	End Time	Name		2026/7/25	2026/7/26	2026/7/27	2026/7/28	2026/7/29	2026/7/30	2026/7/31									
4		2026/6/1	0:00	10:00	Sheryl		0:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
5		2026/6/1	11:00	23:00	Tobby		1:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
6		2026/6/2	0:00	23:00	Tobby		2:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
7		2026/6/3	0:00	23:00	Tobby		3:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
8		2026/6/4	0:00	23:00	Sheryl		4:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
9		2026/6/5	0:00	10:00	Sheryl		5:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
10		2026/6/5	11:00	23:00	Tobby		6:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
11		2026/6/6	0:00	23:00	Tobby		7:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
12		2026/6/7	0:00	23:00	Kenny		8:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
13		2026/6/8	0:00	23:00	Sheryl		9:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
14		2026/6/9	0:00	23:00	Tobby		10:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
15		2026/6/10	0:00	23:00	Tobby		11:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
16		2026/6/11	0:00	23:00	Sheryl		12:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
17		2026/6/12	13:00	23:00	Sheryl		13:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
18		2026/6/12	0:00	12:00	tobby		14:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
19		2026/6/13	0:00	23:00	Kenny		15:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
20		2026/6/14	0:00	23:00	Tobby		16:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
21		2026/6/15	0:00	23:00	Tobby		17:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
22		2026/6/16	0:00	23:00	Tobby		18:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
23		2026/6/17	0:00	23:00	Tobby		19:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
24		2026/6/19	0:00	23:00	Tobby		20:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
25		2026/6/22	11:00	23:00	Sheryl		21:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
26		2026/6/23	0:00	23:00	Sheryl		22:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
27		2026/6/24	19:00	23:00	Henry		23:00		Richard	Tobby	Tobby	Sheryl	Sheryl									
28		2026/6/25	0:00	12:00	Sheryl																	

Reach out to TEEP David

and

Time schedule

Informed of determining the topic

0720

Starting to repulicate Toby's
Experiment

0722

Solve the VPN
Start the first simulation

0723

Discuss with Gu professor

0730

Informed of reaching to David
for the identical topic

0723

Email to David
Seek suggestions from David about
pending topic

0726

Pending works

Pending works

-> This organization comes from my Git.

Priority	Things	Impact	Plan
0	Reading David StudyNotes	The clarity of the topic -> the progress after the new semester	Like replicating Toby's experiment, use same approach to learn on it
1	Do the StudyNotes on github	Check my progress	Post it firstly, Check its quality afterwards
2	progress after the new semester	My GPA and future opportunities	Finish David's StudyNotes first
3	Lab architecture	Resources	Review the files in LINE or talk with the seniors
1	Discuss with prof. Gu	The clarity of the topic and future opportunities	Finish David Studynotes first so that I really understand what I need to do
3	Take course	Future opportunities	Wait for the notice in August on LINE group